

1.2 The structure and functions of the cardio-respiratory system	1.2.1	Functions of the cardiovascular system applied to performance in physical activities: transport of oxygen, carbon dioxide and nutrients, clotting of open wounds, regulation of body temperature
	1.2.2	Structure of the cardiovascular system: atria, ventricles, septum, tricuspid, bicuspid and semi-lunar valves, aorta, vena cava, pulmonary artery, pulmonary vein, and their role in maintaining blood circulation during performance in physical activity and sport
	1.2.3	Structure of arteries, capillaries and veins and how this relates to function and importance during physical activity and sport in terms of blood pressure, oxygenated, deoxygenated blood and changes due to physical exercise

Subject content	What students need to learn
	1.2.4 The mechanisms required (vasoconstriction, vasodilation) and the need for redistribution of blood flow (vascular shunting) during physical activities compared to when resting
	1.2.5 Function and importance of red and white blood cells, platelets and plasma for physical activity and sport
	1.2.6 Composition of inhaled and exhaled air and the impact of physical activity and sport on this composition
	1.2.7 Vital capacity and tidal volume, and change in tidal volume due to physical activity and sport, and the reasons that make the change in tidal volume necessary
	1.2.8 Location of main components of respiratory system (lungs, bronchi, bronchioles, alveoli, diaphragm) and their role in movement of oxygen and carbon dioxide into and out of the body
	1.2.9 Structure of alveoli to enable gas exchange and the process of gas exchange to meet the demands of varying intensities of exercise (aerobic and anaerobic)
	1.2.10 How the cardiovascular and respiratory systems work together to allow participation in physical activity and sport